

Shaz (Ethan Shahrooz) Moghaddam

Data Scientist · Python Developer

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Right to Work: British citizen · DBS Check: Available on request

PROFILE

Imperial College London-trained Data Scientist (96% average) and Python developer who takes projects from raw data to deployed product. Work spans machine learning, NLP, and causal AI, built into production-grade full-stack platforms rather than notebooks. Recent projects include VoltEdge and VoltScope, a pair of B2B energy-intelligence tools combining real-time monitoring, anomaly detection, AI-powered bill extraction, and CSRD-ready ESG reporting, both deployed and backed by automated test suites. Seven Android and Flutter apps shipped to the Google Play Store add end-to-end product ownership and user-driven iteration.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Dart (Flutter), Kotlin & Java, HTML, CSS

Libraries: pandas, NumPy, scikit-learn, spaCy, Pydantic, Matplotlib, Seaborn, SciPy, Plotly, TensorFlow, NetworkX

Frameworks & Tools: FastAPI, Plotly Dash, React, Flask, Streamlit, Electron, Node.js, Firebase, SQLAlchemy, Docker, Git, Jupyter, REST APIs, Power BI, Tableau

Cloud & Deploy: AWS (ECS Fargate), GCP (Cloud Run), Render, Railway, Docker (multi-stage)

Machine Learning: Regression, Classification, Anomaly Detection (Isolation Forest), Causal AI (PC Algorithm), NLP, Random Forest, Gradient Boosting

AI Integration: Anthropic Claude API, ChatGPT, Gemini (integrated into products and development workflows)

Mobile: Android (Kotlin & Java), Flutter (Dart), Google Play Store

SELECTED PROJECTS

VoltEdge — Enterprise Energy Intelligence Platform

FastAPI · Plotly Dash · scikit-learn · Causal AI · SQLite · Docker · Claude API · Render

- Real-time multi-site energy monitoring across 4 industrial sites with 90 days of seeded demo data; per-site ML anomaly detection (Isolation Forest) plus causal AI root-cause analysis (PC algorithm, NetworkX) with quantified causal-strength scores.
- CSRD-ready ESG reporting across Scope 2 emissions, GRI 302-1, and carbon intensity, with a plain-English AI assistant (Claude) for natural-language querying of site data.
- JWT authentication with RBAC, a SOC 2 audit trail, and 903 passing tests; deployed to production on Render via Docker.

GitHub: github.com/ShazMoghaddam/voltedge · Demo: voltedge-1k6q.onrender.com

VoltScope — UK Energy Bill Intelligence & Extraction

FastAPI · Claude API · Pydantic · SQLite · Docker · Render

- Structured extraction and deterministic validation for UK commercial energy bills, and the bill-intelligence module of VoltEdge; sends PDFs and phone photos (including iPhone HEIC) to Claude for schema-constrained extraction into a fixed Pydantic model.
- Registry-based validation engine covering MPAN checks, subtotal and VAT reconciliation, estimated reads, duplicate charges, and contract-renewal windows, plus a cross-portfolio intelligence layer surfacing renewals, quantified suspected overcharges, and supplier comparisons.
- FastAPI review app with drag-and-drop upload, SQLite persistence, API-key auth, and JSON/Excel/CSV export; OpenAPI docs, a Render deployment blueprint, and 50 passing golden-file tests anchored on real anonymised bills.

GitHub: github.com/ShazMoghaddam/voltscope

ClinIQ — Clinical Trial Site Performance Intelligence

FastAPI · Plotly Dash · SQLAlchemy · scikit-learn · spaCy · Claude API · Docker

- B2B platform for monitoring clinical trial site performance across mid-market CROs in the UK and EU, tracking enrolment velocity, risk scoring, and deviation patterns across sites in real time.
- NLP-driven deviation analysis with spaCy, ML risk scoring with scikit-learn, and a plain-English AI assistant (Claude) for natural-language querying of trial data.
- Full-stack production build: FastAPI backend, Plotly Dash frontend, SQLAlchemy ORM, and Docker deployment.

GitHub: github.com/ShazMoghaddam/clinIQ

SalaryAxis — UK Salary Intelligence Platform

Python · Flask · React · pandas · NumPy · SciPy · Docker · AWS / GCP · ONS Open Data

- Production-grade salary benchmarking platform built on ONS ASHE open data, with CPI inflation adjustment, regional breakdowns, gender pay gap trends, occupation distribution profiles, and CAGR analysis.
- REST API with a React dashboard, deployable to AWS ECS Fargate or GCP Cloud Run via multi-stage Docker. GDPR-compliant with no personal data collected.

GitHub: github.com/ShazMoghaddam/salaryaxis

CVLens — AI CV Reader & Analyser

Python · spaCy · NLP · pdfplumber · python-docx

- NLP tool that parses CVs in PDF and DOCX, extracting skills, detecting section structure, and flagging weaknesses through tokenisation, entity recognition, and rule-based matching.

GitHub: github.com/ShazMoghaddam/CVLens

World Map — Interactive Choropleth Explorer

HTML5 · CSS · Vanilla JS · SVG

- Interactive choropleth covering 172 countries across 7 data layers (GDP per capita, HDI, CO₂, life expectancy, and more), with region and political-group filtering, custom data import, shareable URLs, and iframe embedding via a full JS API. Single HTML file, zero dependencies, MIT licensed.

GitHub: github.com/ShazMoghaddam/world-map · Demo: shazmoghaddam.github.io/world-map

EXPERIENCE

Independent Product & Software Developer

2022 – Present

Remote · Google Play Store · Gumroad / Lemon Squeezy / Envato

- Built and shipped seven Android and Flutter apps to the Google Play Store, owning the full product lifecycle from concept to release and iteration.
- Maintain a digital-product catalogue across Gumroad, Lemon Squeezy, and Envato, spanning finance tools (Market Pulse, TradeHours Pro), generative art systems (Typography Digits, KNOK Matrix), and screensavers (Aero Atlas), iterating on features from user feedback.

Sales Representative (Technology)

Mar 2011 – Mar 2013

John Lewis / Currys (Agency) · London

- Advised customers on technical products across multiple London sites, building communication, prioritisation, and stakeholder-management skills.

EDUCATION & CERTIFICATIONS

Data Science Bootcamp — Imperial College London / HyperionDev · June 2025 · 96% average

Machine learning, NLP, statistical analysis, data cleaning, and visualisation.

Google Play Store Listing Certificate — Google Play Academy · April 2026 (valid through April 2029)

The Data Science Course: Complete Data Science Bootcamp — Udemy · October 2025 · 32 hours

iOS & Swift: The Complete iOS App Development Bootcamp — Udemy · January 2024 · 60.5 hours

100 Days of Code & The Complete Python Bootcamp — Udemy · Oct 2022 – Jul 2023 · 58 hours

ADDITIONAL

Writing: Seven personal essays on Medium covering growth, identity, technology, and perspective (medium.com/@shaz.moghaddam).

Available for: Full-time · Freelance / Contract · Remote · Hybrid · Open to international opportunities.